Physical Chemistry (I)

Lecture 21

Chemical Equilibria

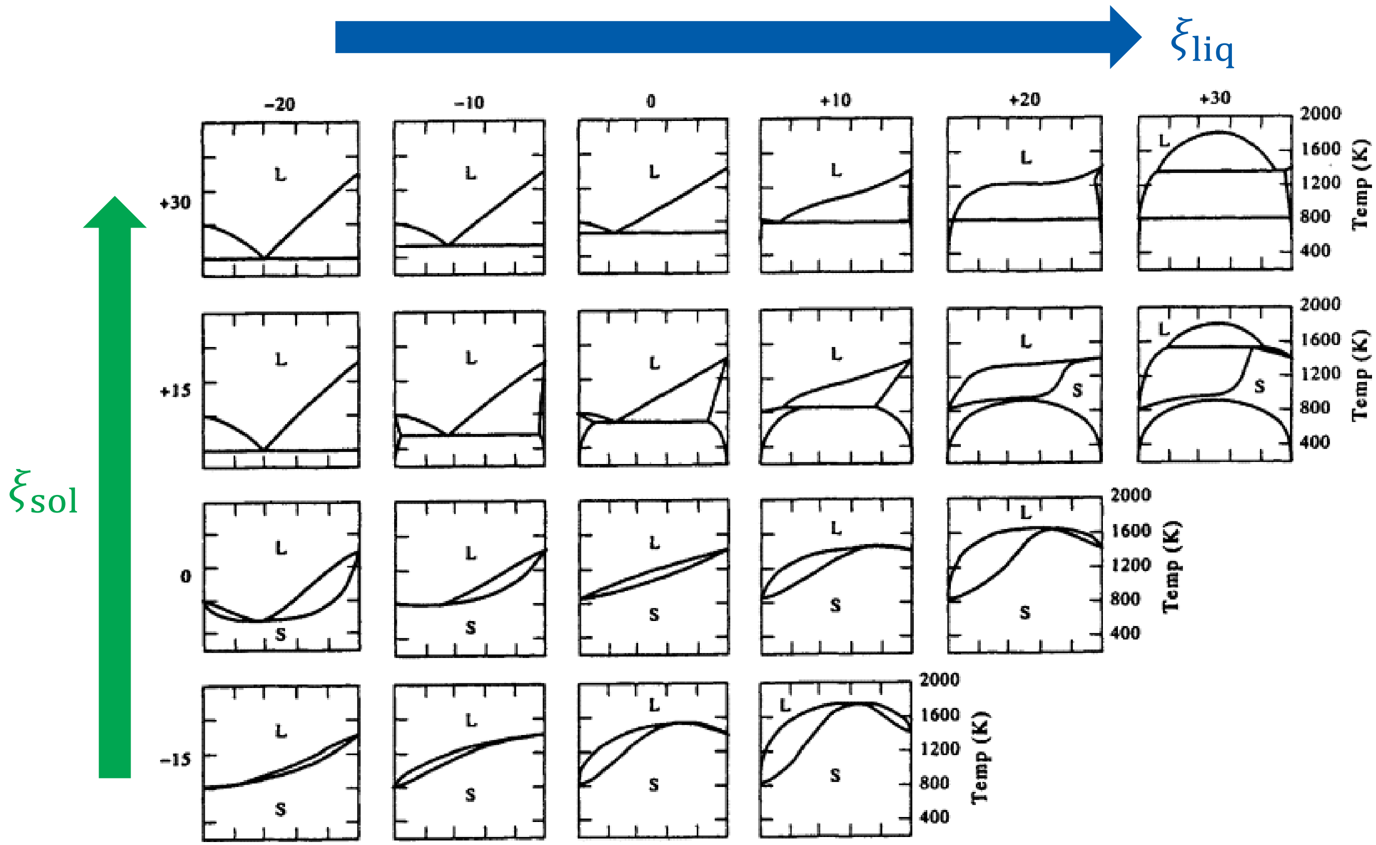
In the Last Class

- Phase diagrams of liquid-solid mixtures
  - Simple systems
  - Reacting systems

- Colligative properties
  - Thermodynamic origin
  - Derivations

Last class

Liquid-Solid Phase Diagrams

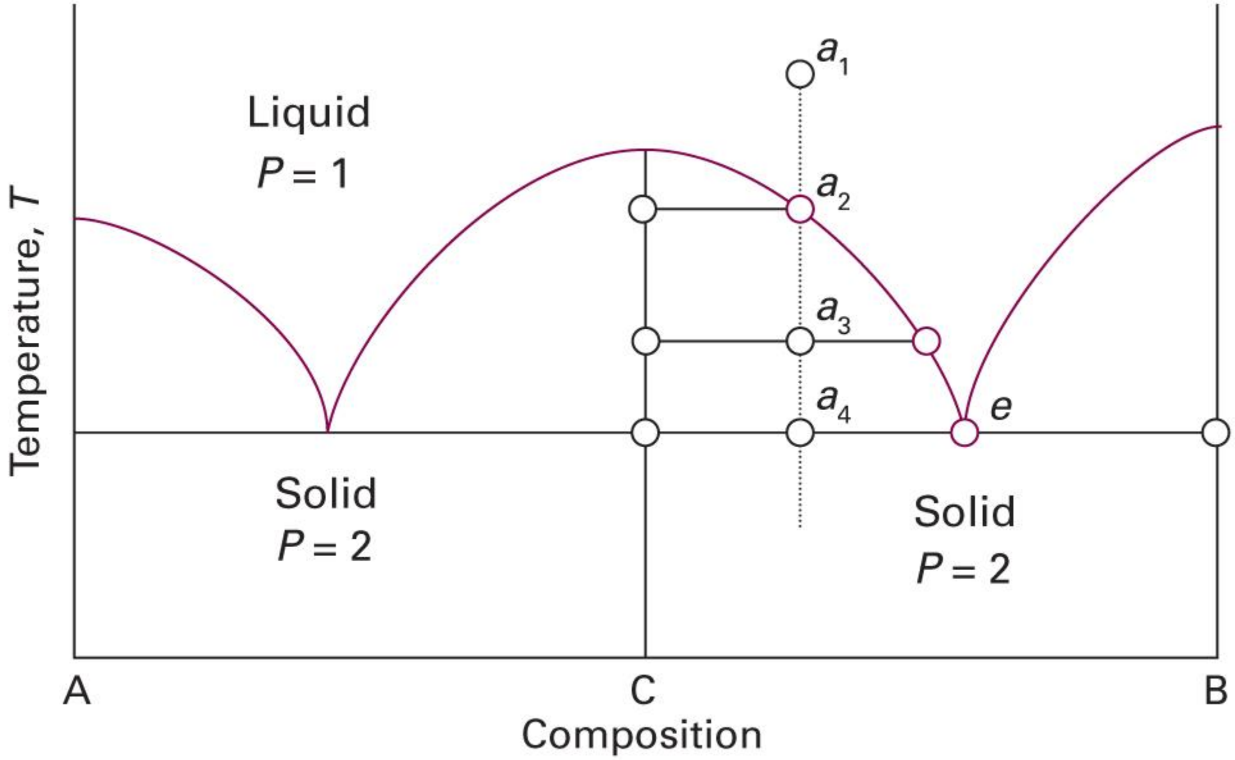


(Image: Gaskell, *Introduction to the Thermodynamics of Materials*, 5th Ed., Fig. 10.23)

Last class

Congruent Melting

(assuming completely immiscible solid)



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5D.4)

Last class

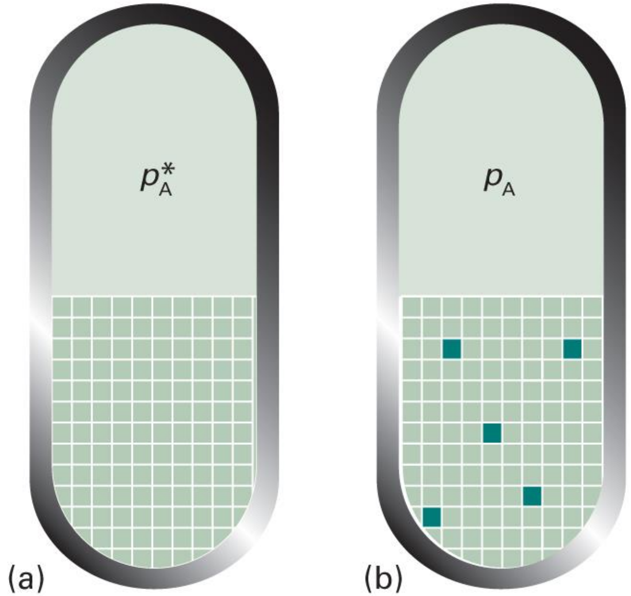
Colligative Properties

- Physical properties that depends on the relative number of solute particles present but not their chemical identity

1. Lowering of vapor pressure
2. Elevation of boiling point
3. Depression of freezing point
4. Osmotic pressure

Origin of Colligative Properties

- Due to the reduction of the chemical potential of the liquid solvent as a result of the presence of solute

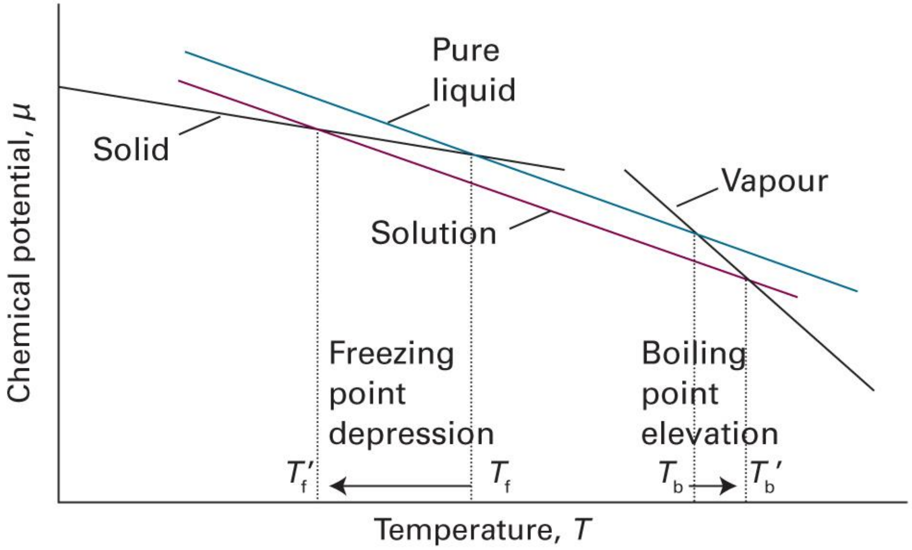


(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5B.7)

Last class

Changes in Phase Transitions

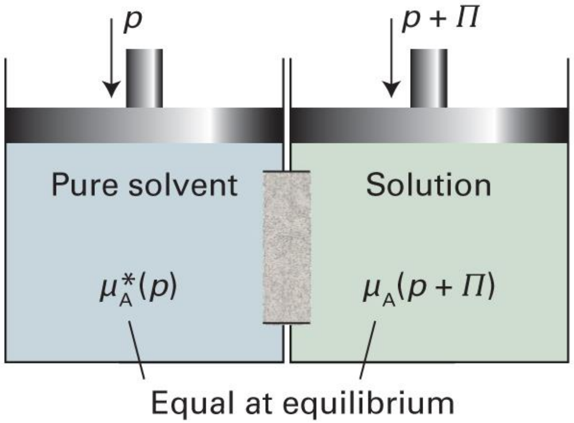
$$\mu_A = \mu_A^* + RT \ln x_A \ (x_A < 1)$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5B.6)

Osmosis

the spontaneous passage of a pure solvent into a solution separated from it by a semipermeable membrane



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5B.12)

van’t Hoff Factor

e.g. $NaCl(s) \longrightarrow Na^{+}(aq) + Cl^{-}(aq)$

- van’t Hoff factor i

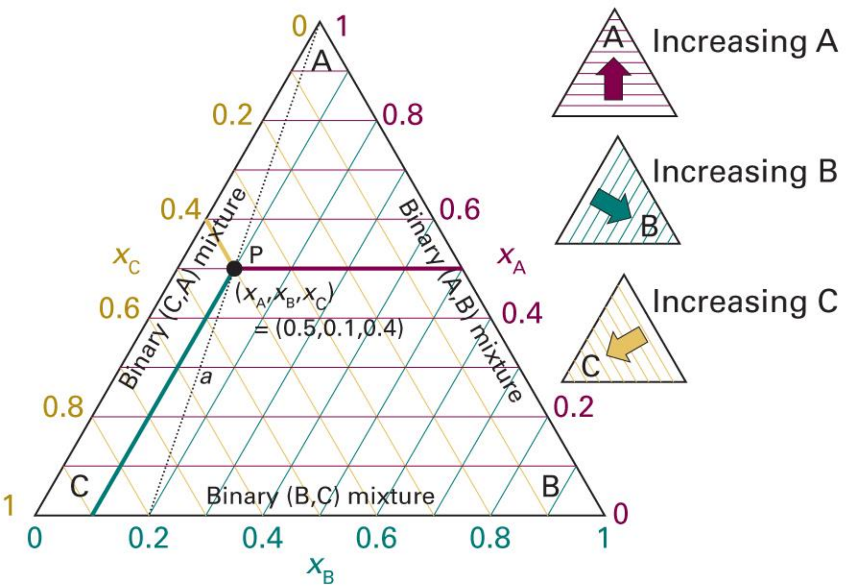
$$\Delta T_{\mathrm{b}} = iK_{\mathrm{b}}b$$

$$\Delta T_{\mathrm{f}} = iK_{\mathrm{f}}b$$

$$\Pi = i[\mathrm{B}]RT$$

Ternary Systems: Triangle

$$x_{\mathrm{A}} + x_{\mathrm{B}} + x_{\mathrm{C}} = 1$$

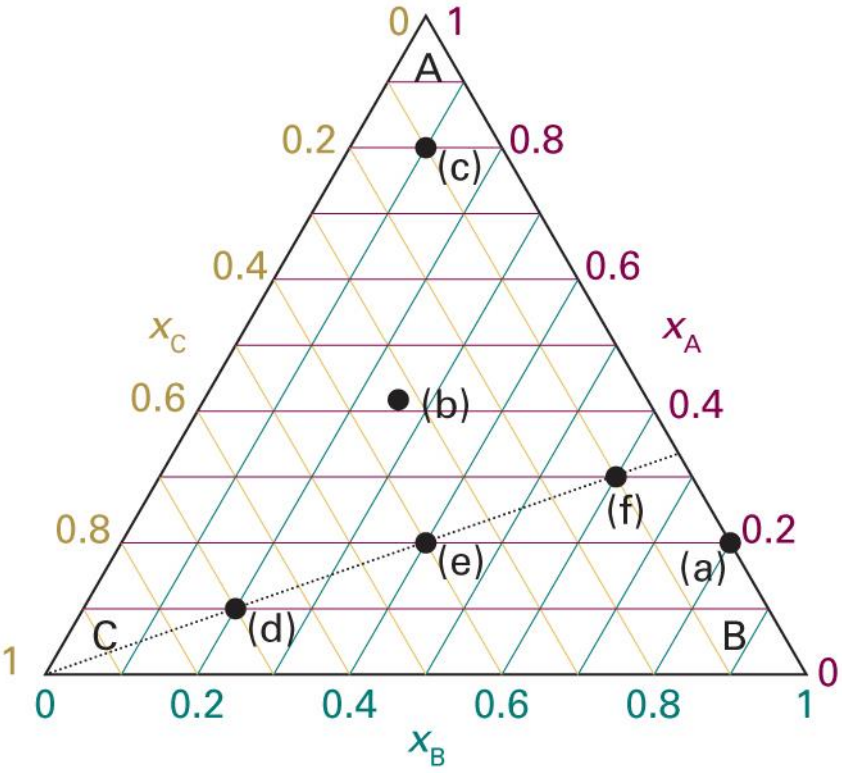


(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5E.1)

Brief Illustration 5E.1

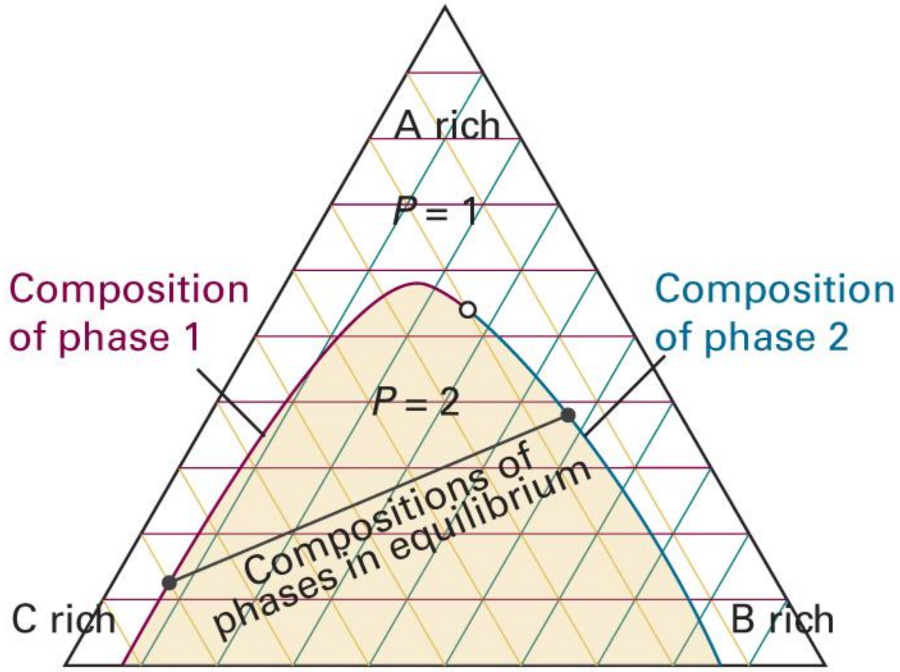
Indicate the following points:

Point	x_A	x_B	x_C
a	0.20	0.80	0
b	0.42	0.26	0.32
c	0.80	0.10	0.10
d	0.10	0.20	0.70
e	0.20	0.40	0.40
f	0.30	0.60	0.10



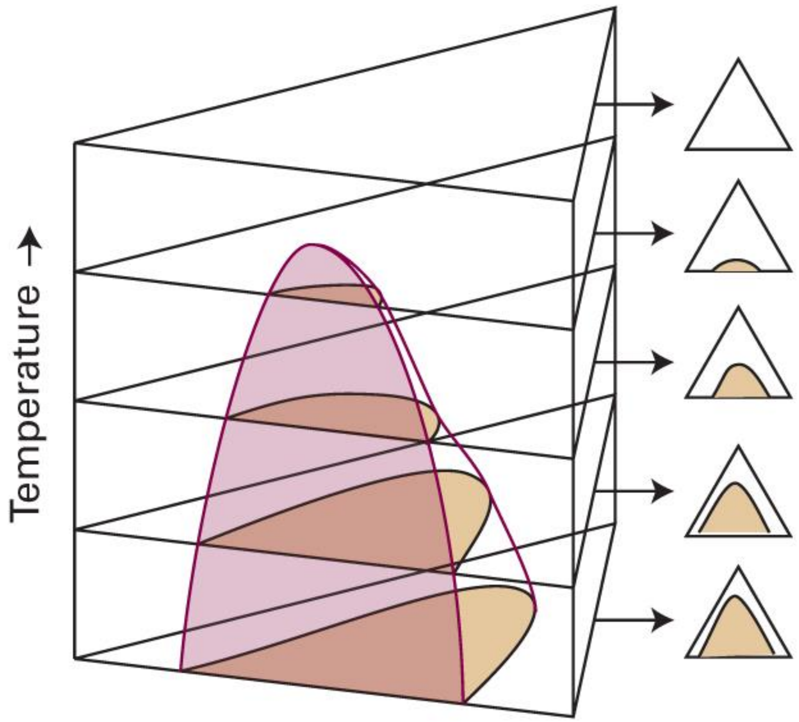
(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5E.2)

Partially Miscible Liquids



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5E.5)

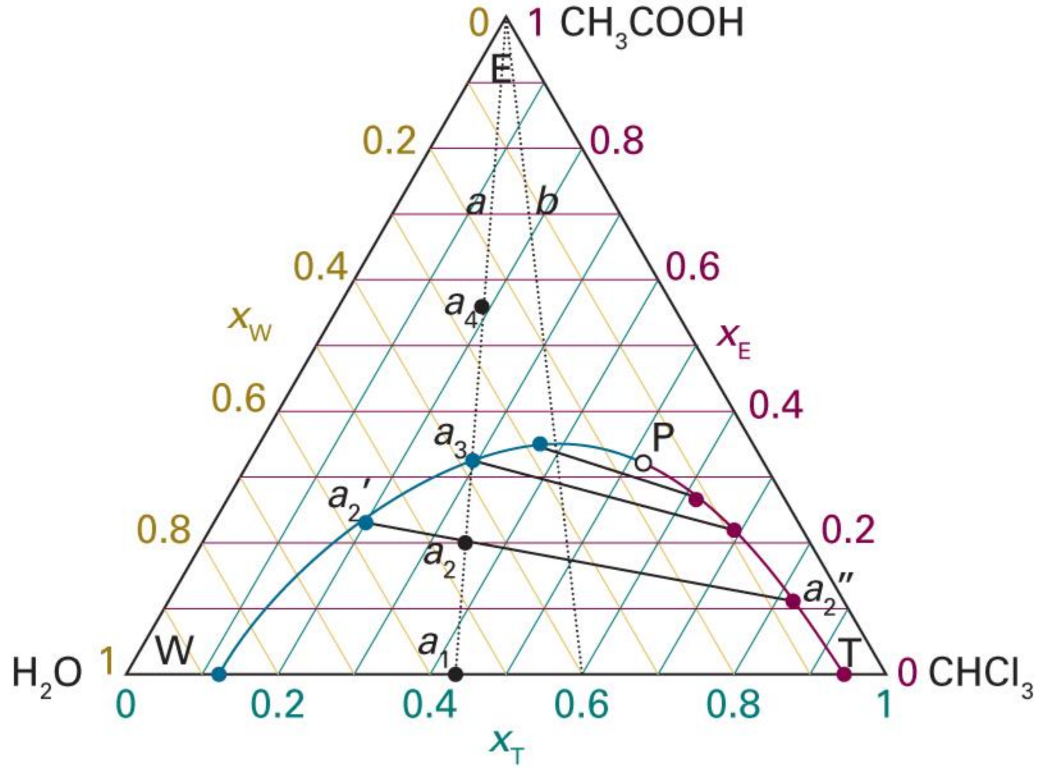
What If Temperature Changes?



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5E.3)

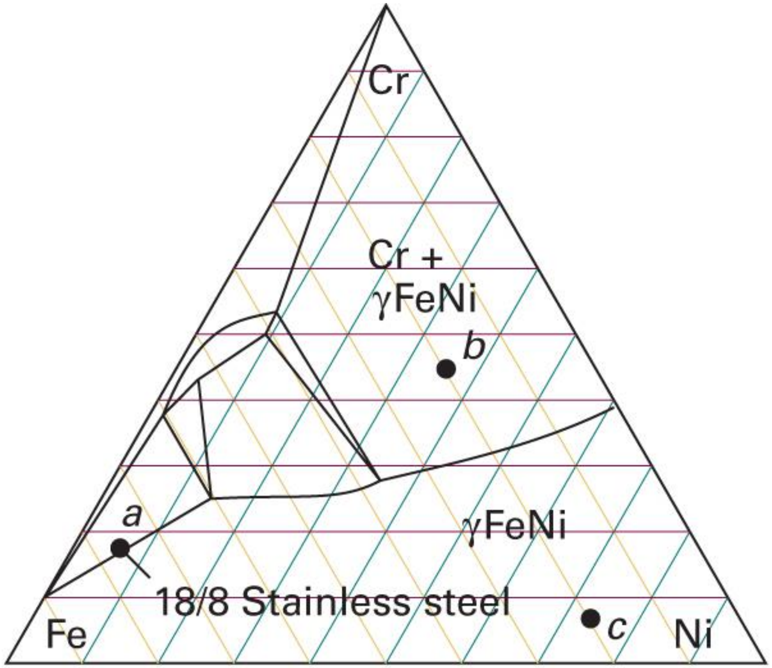
Topic 5E

Example: CH_3COOH-H_2O-$CHCl_3$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5E.4)

Ternary Solids



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5E.6)

Summary

- Phase
- Chemical potential
- Gibbs-Duhem eq.

Multi-component system

Gas mixture

- Dalton's law
- Perfect gas
- Real gas

Ternary systems

Liquid mixture

Ideal solution

- Raoult's law

$$p_{\mathrm{A}} = x_{\mathrm{A}} p_{\mathrm{A}}^{*}$$

- Chemical potential

$$\mu_{\mathrm{A}} = \mu_{\mathrm{A}}^{*} + RT \ln x_{\mathrm{A}}$$

- $\Delta_{\mathrm{mix}} G = -T \Delta_{\mathrm{mix}} S$

Ideal-dilute solution

- Henry's law

$$p_{\mathrm{B}} = x_{\mathrm{B}} K_{\mathrm{B}}$$

- Chemical potential

$$\mu_{\mathrm{B}} = \mu_{\mathrm{B}}^{\ominus} + RT \ln x_{\mathrm{B}}$$

Real solution

- Regular solution
- Bragg-Williams model
- Excess function
- Activity

$$\mu_{\mathrm{A}} = \mu_{\mathrm{A}}^{*} + RT \ln a_{\mathrm{A}}$$

$$\mu_{\mathrm{B}} = \mu_{\mathrm{B}}^{\ominus} + RT \ln a_{\mathrm{B}}$$

- L-G phase diagram
- L-L phase diagram
- L-S phase diagram

Colligative properties

9. Chemical Equilibria

Chapter Overview

- Chemical reactions and equilibria
  - Reaction quotient and equilibrium constant
  - Activity approximations
  - Gaseous reactions
- Le Chatelier's principle

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Chemical Reactions

$$aA + bB \rightleftarrows cC + dD$$

Can we construct

thermodynamics of chemical reactions?

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Extent of Reaction ξ

$$a\text{A} + b\text{B} \rightleftarrows c\text{C} + d\text{D}$$

Species	Initial	→	Change	→	Final
A	n_A		$-a\xi$		$n_\text{A} - a\xi$
B	n_B		$-b\xi$		$n_\text{B} - b\xi$
C	n_C		$+c\xi$		$n_\text{C} + c\xi$
D	n_D		$+d\xi$		$n_\text{D} + d\xi$

Simple Reaction

$A(g) \rightleftarrows B(g)$

- Extent of reaction ξ: for $\mathrm{d}\xi$, $\mathrm{d}n_\mathrm{A} = -\mathrm{d}\xi$ and $\mathrm{d}n_\mathrm{B} = +\mathrm{d}\xi$
- Gibbs energy $G = G(p, T, n_\mathrm{A}, n_\mathrm{B})$

  For constant p and T,

$$\mathrm{d}G = \mu_\mathrm{A}\mathrm{d}n_\mathrm{A} + \mu_\mathrm{B}\mathrm{d}n_\mathrm{B} = -\mu_\mathrm{A}\mathrm{d}\xi + \mu_\mathrm{B}\mathrm{d}\xi = (\mu_\mathrm{B} - \mu_\mathrm{A})\mathrm{d}\xi$$

$$\left(\frac{\partial G}{\partial \xi}\right)_{p,T} = \mu_\mathrm{B} - \mu_\mathrm{A}$$

Reaction Gibbs Energy

$$A(g) \rightleftarrows B(g)$$

Assuming the absolute scale,

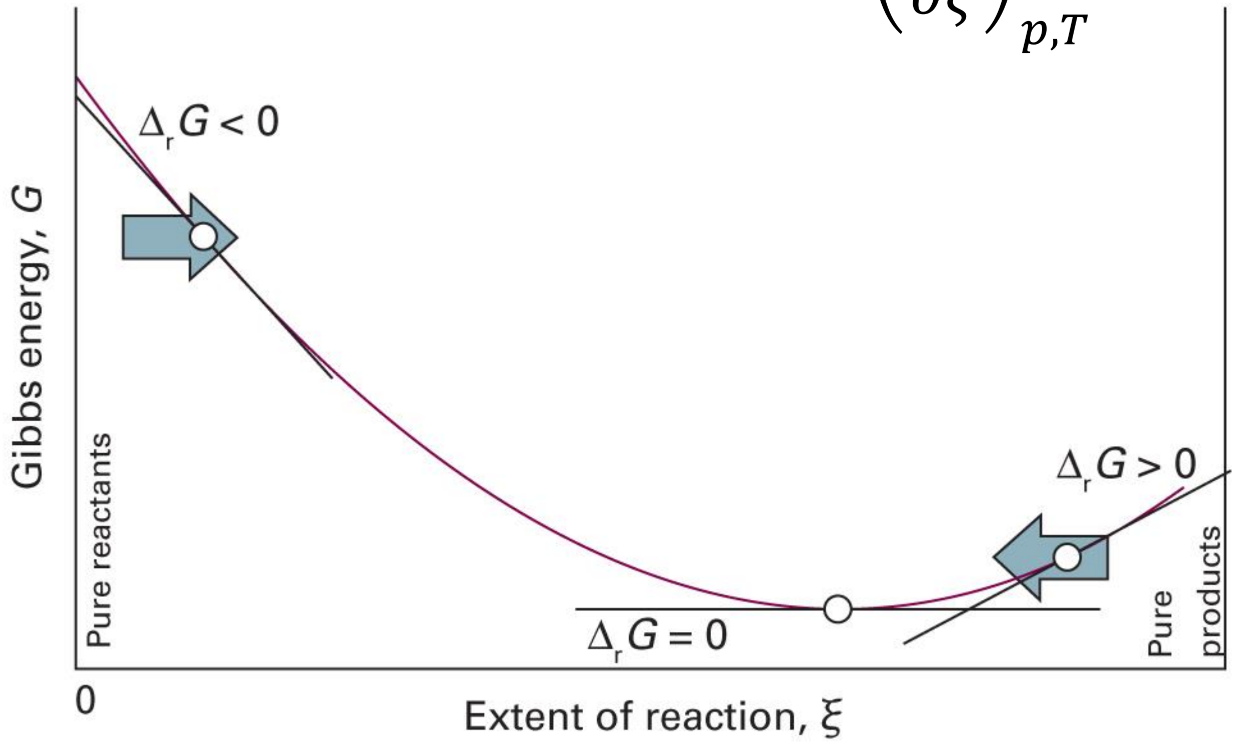
$$\Delta_r G = G_{m,B} - G_{m,A} = \mu_B - \mu_A = \left(\frac{\partial G}{\partial \xi}\right)_{p,T}$$

- $\Delta_r G = 0$: reversible, equilibrium
- $\Delta_r G < 0$: irreversible, spontaneous reaction
- $\Delta_r G > 0$: impossible $\rightarrow$ spontaneous reverse reaction

Topic 6A

Progress of Reaction

$$G = \mu_\mathrm{A} n_\mathrm{A} + \mu_\mathrm{B} n_\mathrm{B} \qquad \Delta_\mathrm{r} G = \left(\frac{\partial G}{\partial \xi}\right)_{p,T}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 6A.1)

Reaction Quotient

A(g) ⇄ B(g)

For a perfect gas, $\mu = \mu^{\ominus} + RT\ln(p/p^{\ominus})$

$$\Delta_r G = \mu_B - \mu_A = \left\{\mu_B^{\ominus} + RT\ln\left(\frac{p_B}{p_B^{\ominus}}\right)\right\} - \left\{\mu_A^{\ominus} + RT\ln\left(\frac{p_A}{p_A^{\ominus}}\right)\right\}$$

Hence, using $\Delta_r G^{\ominus} = \mu_B^{\ominus} - \mu_A^{\ominus}$,

$$\Delta_r G = \Delta_r G^{\ominus} + RT\ln\left(\frac{p_B}{p_A}\right) = \Delta_r G^{\ominus} + RT\ln Q$$

Equilibrium Constant

$$\Delta_r G = \Delta_r G^{\ominus} + RT \ln Q$$

At equilibrium, $\Delta_r G = 0$

$$0 = \Delta_r G^{\ominus} + RT \ln Q_{\mathrm{eq}} = \Delta_r G^{\ominus} + RT \ln K$$

Hence,

$$\Delta_r G^{\ominus} = -RT \ln K$$, where $K = (p_{\mathrm{B}}/p_{\mathrm{A}})_{\mathrm{eq}}$

Generalization

For an arbitrary reaction,

$$\sum\nolimits_{\text{reactant } i} \nu_i \, \{i\} \rightleftarrows \sum\nolimits_{\text{product } j} \nu_j \, \{j\}$$

e.g. 2A + B $\rightleftarrows$ 3C + D

$\nu_{\mathrm{A}} = 2$, $\nu_{\mathrm{B}} = 1$, $\nu_{\mathrm{C}} = 3$, $\nu_{\mathrm{D}} = 1$

Now,

$$\mathrm{d}G = \sum\nolimits_{\text{product } j} \mu_j \mathrm{d}n_j + \sum\nolimits_{\text{reactant } i} \mu_i \mathrm{d}n_i$$

Generalization

$$\mathrm{d}G = \sum\nolimits_{\text{product } j} \mu_j \mathrm{d}n_j + \sum\nolimits_{\text{reactant } i} \mu_i \mathrm{d}n_i$$

$$\mathrm{d}G = \sum\nolimits_{\text{product } j} \mu_j \nu_j \mathrm{d}\xi - \sum\nolimits_{\text{reactant } i} \mu_i \nu_i \mathrm{d}\xi$$

$$\mathrm{d}G = \left(\sum\nolimits_{\text{product } j} \mu_j \nu_j - \sum\nolimits_{\text{reactant } i} \mu_i \nu_i \right) \mathrm{d}\xi$$

$$\Delta_{\mathrm{r}} G = \left(\frac{\partial G}{\partial \xi} \right)_{p,T} = \sum\nolimits_{\text{product } j} \mu_j \nu_j - \sum\nolimits_{\text{reactant } i} \mu_i \nu_i$$

Topic 6A

Chemical Potential and Activity

- Real gas: $\mu = \mu^{\ominus} + RT\ln(f/p^{\ominus})$
- Real solution (solvent): $\mu = \mu^{*} + RT\ln a$
- Real solution (solute): $\mu = \mu^{\ominus} + RT\ln a$

- Generally, let us use

$$\mu = \mu^{\ominus} + RT\ln a$$

Generalization

$$\Delta_r G = \sum_{\text{product } j} \nu_j \mu_j - \sum_{\text{reactant } i} \nu_i \mu_i$$

For each species x,

$$\nu_x \mu_x = \nu_x (\mu_x^{\ominus} + RT \ln a_x) = \nu_x \mu_x^{\ominus} + RT \ln\left(a_x^{\nu_x}\right)$$

Then

$$\sum_x \nu_x \mu_x = \sum_x \nu_x \mu_x^{\ominus} + RT \sum_x \ln\left(a_x^{\nu_x}\right)$$

$$= \sum_x \nu_x \mu_x^{\ominus} + RT \ln \prod_x a_x^{\nu_x}$$

Generalization

$$\Delta_r G = \sum\nolimits_{\text{product } j} \nu_j \mu_j - \sum\nolimits_{\text{reactant } i} \nu_i \mu_i$$

Substitution leads to

$$\Delta_r G = \Delta_r G^{\ominus} + RT \ln \prod_{\text{product } j} a_j^{\nu_j} - RT \ln \prod_{\text{reactant } i} a_i^{\nu_i}$$

where $\Delta_r G^{\ominus} = \sum_{\text{product } j} \nu_j \mu_j^{\ominus} - \sum_{\text{reactant } i} \nu_i \mu_i^{\ominus}$

$$\Delta_r G = \Delta_r G^{\ominus} + RT \ln \left(\prod_{\text{product } j} a_j^{\nu_j} \Big/ \prod_{\text{reactant } i} a_i^{\nu_i} \right)$$

Q and K

$$\Delta_{\mathrm{r}}G = \Delta_{\mathrm{r}}G^{\ominus} + RT\ln Q$$

where the reaction quotient Q is defined as

$$Q = \frac{\prod_{\mathrm{product}\ j} a_j^{\nu_j}}{\prod_{\mathrm{reactant}\ i} a_i^{\nu_i}}$$

At equilibrium, $\Delta_{\mathrm{r}}G = 0$

$$0 = \Delta_{\mathrm{r}}G^{\ominus} + RT\ln Q_{\mathrm{eq}} = \Delta_{\mathrm{r}}G^{\ominus} + RT\ln K$$

where the equilibrium constant K is Q at equilibrium

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What is $\Delta_\mathrm{r} G^{\ominus}$?

$$\Delta_\mathrm{r} G^{\ominus} = \sum_{\text{products } j} \nu_j \mu_j^{\ominus} - \sum_{\text{reactants } i} \nu_i \mu_i^{\ominus}$$

Here, $\mu_x^{\ominus}$ is the chemical potential of species x ...

- Gas: at standard pressure (1 bar)
- Solute/solvent: in its pure form

Thus, $\Delta_\mathrm{r} G^{\ominus}$ is the standard Gibbs energy of reaction.

Connection to Thermochemistry

$$\Delta_r G^{\ominus} = -RT \ln K$$

Since we know

$$\Delta_r G^{\ominus} = \sum_{\text{products } j} \nu_j \Delta_f G^{\ominus}(j) - \sum_{\text{reactants } i} \nu_i \Delta_f G^{\ominus}(i)\,,$$

we can calculate K from thermochemical data.

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Meaning of Q and K

- Reaction quotient Q
  - "How far the equilibrium is"
  - Varies as the reaction progresses
  - $Q = K$ at equilibrium

- Equilibrium constant K
  - Amounts of reactants and products at equilibrium
  - Constant for the given experimental condition

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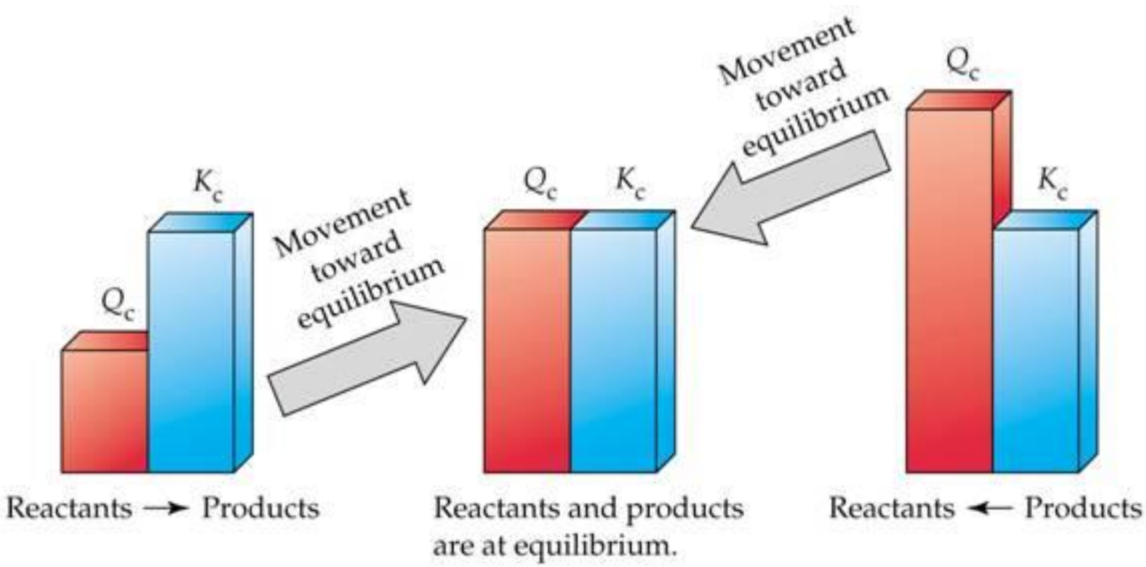
Meaning of Q and K



(Image: http://www.fccj.us/chm2046/SampleTest/46M10dAnswers.htm)

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Activity

$$\mu = \mu^{\ominus} + RT \ln a$$

- Gas: $\mu = \mu^{\ominus} + RT \ln(f/p^{\ominus})$
- Solution
  - Solvent: $\mu_A = \mu_A^* + RT \ln a_A$
  - Solute: $\mu_B = \mu_B^* + RT \ln(K_B/p_B^*) + RT \ln a_B$
- Pure liquid and solid: ?

Approximation (perfect/ideal limit)?

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Approximate Activity of Gas

$$\mu = \mu^{\ominus} + RT \ln a = \mu^{\ominus} + RT \ln(f/p^{\ominus})$$

For a perfect gas,

$$\mu = \mu^{\ominus} + RT \ln(p/p^{\ominus})$$

Hence, approximately

$$a = \frac{p}{p^{\ominus}}$$

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Approximate Activity of Solute

$$\mu = \mu^{\ominus} + RT \ln a$$

For an ideal-dilute solution,

$$\mu_B = \mu_B^* + RT \ln(K_B/p_B^*) + RT \ln x_B$$

$$x_B = \frac{n_B}{n_A + n_B} \approx \frac{n_B}{n_A} = \frac{V}{n_A}\frac{n_B}{V} = \frac{[B]}{[A]} = \frac{[B]}{c^{\ominus}}\frac{c^{\ominus}}{[A]}$$

Assuming that [A] is constant,

$$\mu_B = \underbrace{\mu_B^* + RT \ln(K_B/p_B^*) - RT \ln([A]/c^{\ominus})}_{\mu_B^{\ominus}} + RT \ln\underbrace{([B]/c^{\ominus})}_{a}$$

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Approximate Activity of Solvent

$$\mu = \mu^{\ominus} + RT \ln a$$

For an ideal-dilute solution,

$$\mu_A = \mu_A^* + RT \ln x_A$$

$$x_A = \frac{n_A}{n_A + n_B} \approx \frac{n_A}{n_A} = 1$$

Hence, approximately

$$a = 1$$

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Pure Solids and Liquids

$$\Delta G_\mathrm{m} = \mu_f - \mu_i = \int_{p_i}^{p_f} V_\mathrm{m}\, \mathrm{d}p$$

Let the initial state to be the standard state:

$$\mu - \mu^\ominus = \int_{p^\ominus}^{p} V_\mathrm{m}\, \mathrm{d}p' \approx V_\mathrm{m} \int_{p^\ominus}^{p} 1\, \mathrm{d}p'$$

From the definition of an activity, $\mu - \mu^\ominus = RT \ln a$,

$$\ln a = \frac{V_\mathrm{m}}{RT} \int_{p^\ominus}^{p} 1\, \mathrm{d}p' = \frac{V_\mathrm{m}}{RT} (p - p^\ominus)$$

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Example: Activity of Coke

The density of coke at 1000°C is about 1.5 g cm^{-3}. Calculate the activity of C(s) in the form of coke at 100 bar and 1000°C.

The molar volume is

V_m = (molar mass)/(density) = (12 g/mol)/(1.5 g cm^{-3}) = 8.0 cm^3 mol^{-1}

Thus,

$$\ln a = V_m(p - p^\ominus)/RT$$

$$= (8.0\ \text{cm}^3\ \text{mol}^{-1})\times(100 - 1)\ \text{bar}/(0.082\ \text{L bar K}^{-1}\ \text{mol}^{-1})/(1273\ \text{K})$$

$$= 0.0076$$

or a = 1.01.

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Pure Solids and Liquids

$$\ln a = \frac{V_{\mathrm{m}}}{RT}(p - p^{\ominus})$$

At moderate pressures, the activity is very close to 1.

Hence, we usually take $a = 1$ for pure solids and liquids.

Topic 6A

Approximation for Activity

State	Measure	Approximation for a_J	Definition
Solute	molality	$b_J/b^\ominus$	$b^\ominus = 1\,\text{mol kg}^{-1}$
	molar concentration	$[J]/c^\ominus$	$c^\ominus = 1\,\text{mol dm}^{-3}$
Gas phase	partial pressure	$p_J/p^\ominus$	$p^\ominus = 1\,\text{bar}$
Pure solid, liquid		1 (exact)	

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., p. 210)

For Gaseous Reactions

$$K_p = \frac{\prod_{\text{product } j} a_j^{\nu_j}}{\prod_{\text{reactant } i} a_i^{\nu_i}} = \frac{\prod_{\text{product } j} (p_j/p^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} (p_i/p^\ominus)^{\nu_j}}$$

Since $p_i/p^\ominus = [i](RT/p^\ominus) = [i]/c^\ominus \, (c^\ominus RT/p^\ominus)$,

$$K_p = \frac{\prod_{\text{product } j} ([j]/c^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} ([i]/c^\ominus)^{\nu_i}} \left(\frac{c^\ominus RT}{p^\ominus}\right)^{\sum_j \nu_j - \sum_i \nu_i}$$

$$K_p = K_c \left(\frac{c^\ominus RT}{p^\ominus}\right)^{\Delta\nu} = K_c \left(\frac{T}{12.03\ \mathrm{K}}\right)^{\Delta\nu}$$

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Equilibrium Constants

For gaseous reactions (assuming perfect gases),

$$K_p = \frac{\prod_{\text{product } j} (p_j / p^{\ominus})^{\nu_j}}{\prod_{\text{reactant } i} (p_i / p^{\ominus})^{\nu_j}}$$

$$K_c = \frac{\prod_{\text{product } j} ([j] / c^{\ominus})^{\nu_j}}{\prod_{\text{reactant } i} ([i] / c^{\ominus})^{\nu_i}}$$

Can you theoretically calculate their values?

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Statistical Mechanics

$$\mu = \left(\frac{\partial A}{\partial n}\right)_{T,V} = -k_{\mathrm{B}}T\left(\frac{\partial \ln Z}{\partial n}\right)_{T,V} = -RT\left(\frac{\partial \ln Z}{\partial N}\right)_{T,V}$$

For N perfect-gas molecules, $Z = \frac{1}{N!}z^N$ and

$$\ln Z \approx N\ln z - N\ln N + N$$

Hence,

$$\mu = -RT\ln z + RT\ln N + RT - RT = -RT\ln\left(\frac{z}{N}\right)$$

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Partition Functions in Action

$$\mu_x = -RT \ln\left(\frac{z_x}{N_x}\right)$$

At equilibrium,

$$\Delta_r G = \sum\nolimits_{\text{product } j} \nu_j \mu_j - \sum\nolimits_{\text{reactant } i} \nu_i \mu_i = 0$$

Substitution leads to

$$\sum\nolimits_{\text{product } j} \nu_j \ln\left(\frac{z_j}{N_j}\right) = \sum\nolimits_{\text{reactant } i} \nu_i \ln\left(\frac{z_i}{N_i}\right)$$

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Tedious Math

$$\sum_{\text{product } j} \nu_j \ln\left(\frac{z_j}{N_j}\right) = \sum_{\text{reactant } i} \nu_i \ln\left(\frac{z_i}{N_i}\right)$$

$$\sum_{\text{product } j} \ln\left(\frac{z_j}{N_j}\right)^{\nu_j} = \sum_{\text{reactant } i} \ln\left(\frac{z_i}{N_i}\right)^{\nu_i}$$

$$\prod_{\text{product } j} \left(\frac{z_j}{N_j}\right)^{\nu_j} = \prod_{\text{reactant } i} \left(\frac{z_i}{N_i}\right)^{\nu_i}$$

$$\frac{\prod_{\text{product } j} N_j^{\nu_j}}{\prod_{\text{reactant } i} N_i^{\nu_i}} = \frac{\prod_{\text{product } j} z_j^{\nu_j}}{\prod_{\text{reactant } i} z_i^{\nu_i}}$$

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Equilibrium Constant

$$\frac{\prod_{\text{product } j} N_j^{\nu_j}}{\prod_{\text{reactant } i} N_i^{\nu_i}} = \frac{\prod_{\text{product } j} z_j^{\nu_j}}{\prod_{\text{reactant } i} z_i^{\nu_i}}$$

$$\frac{\prod_{\text{product } j}\{N_j/(N_\mathrm{A}Vc^{\ominus})\}^{\nu_j}}{\prod_{\text{reactant } i}\{N_i/(N_\mathrm{A}Vc^{\ominus})\}^{\nu_i}} = \frac{\prod_{\text{product } j}\{z_j/(N_\mathrm{A}Vc^{\ominus})\}^{\nu_j}}{\prod_{\text{reactant } i}\{z_i/(N_\mathrm{A}Vc^{\ominus})\}^{\nu_i}}$$

The left-hand side is the equilibrium constant:

$$K_c = \frac{\prod_{\text{product } j}([j]/c^{\ominus})^{\nu_j}}{\prod_{\text{reactant } i}([i]/c^{\ominus})^{\nu_i}} = \frac{\prod_{\text{product } j}\{z_j/(N_\mathrm{A}Vc^{\ominus})\}^{\nu_j}}{\prod_{\text{reactant } i}\{z_i/(N_\mathrm{A}Vc^{\ominus})\}^{\nu_i}}$$

Le Chatelier's Principle

When subjected to a disturbance,

a system at equilibrium tends to respond in a way

that minimizes the effect of the disturbance.

Disturbances to consider:

- Amount
- Volume & pressure (for a gaseous reaction)
- Temperature

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Amount Change

- Increase of the amount of a particular species
  - The system responds to decrease the amount
- Decrease of the amount of a particular species
  - The system responds to increase the amount

- Reactant increase / product decrease: $Q < K$
  - Decrease of reactants & increase of products
- Reactant decrease / product increase: $Q > K$
  - Increase of reactants & decrease of products

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Volume/Pressure Change

For a gaseous reaction,

- Increase of the volume → decrease of the pressure
- Decrease of the volume → increase of the pressure
- Pressure change due to addition of inert gas does not affect the equilibrium

$$K_p = \frac{\prod_{\text{product } j} \left(p_j / p^{\ominus}\right)^{\nu_j}}{\prod_{\text{reactant } i} \left(p_i / p^{\ominus}\right)^{\nu_i}} = \frac{\prod_{\text{product } j} \left(n_j RT / V p^{\ominus}\right)^{\nu_j}}{\prod_{\text{reactant } i} \left(n_i RT / V p^{\ominus}\right)^{\nu_i}}$$

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Volume/Pressure Change

$$K_p = \frac{\prod_{\text{product } j}\left(n_j RT / V p^{\ominus}\right)^{\nu_j}}{\prod_{\text{reactant } i}\left(n_i RT / V p^{\ominus}\right)^{\nu_i}}$$

For a volume change from V to xV,

$$Q_p = \frac{\prod_{\text{product } j}\left(n_j RT / xV p^{\ominus}\right)^{\nu_j}}{\prod_{\text{reactant } i}\left(n_i RT / xV p^{\ominus}\right)^{\nu_i}}$$

$$Q_p = K_p x^{-\left(\sum_j \nu_j - \sum_i \nu_i\right)} = K_p x^{-\Delta\nu}$$

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Volume/Pressure Change

$$Q_p = K_p x^{-\Delta\nu}$$

For $x > 1$ (volume increase),

- $\Delta\nu = 0$: $Q_p = K_p$ and the system is at equilibrium
- $\Delta\nu > 0$: $Q_p < K_p$ and the forward reaction is favored
- $\Delta\nu < 0$: $Q_p > K_p$ and the reverse reaction is favored

Temperature Change

- Exothermic reactions
  - Increased temperature favors the reactants
  - Decreased temperature favors the products

- Endothermic reactions
  - Increased temperature favors the products
  - Decreased temperature favors the reactants

van’t Hoff Equation

$$\ln K = -\Delta_r G^{\ominus}/RT$$

Differentiating both sides with respect to T,

$$\frac{\mathrm{d}\ln K}{\mathrm{d}T} = -\frac{1}{R}\frac{\mathrm{d}(\Delta_r G^{\ominus}/T)}{\mathrm{d}T}$$

Using the Gibbs-Helmholtz equation,

$$-R\frac{\mathrm{d}\ln K}{\mathrm{d}T} = \frac{\mathrm{d}(\Delta_r G^{\ominus}/T)}{\mathrm{d}T} = -\frac{\Delta_r H^{\ominus}}{T^2}$$

$$\frac{\mathrm{d}\ln K}{\mathrm{d}T} = \frac{\Delta_r H^{\ominus}}{RT^2}$$

T-Dependence of K

$$\frac{\mathrm{d}\ln K}{\mathrm{d}T} = \frac{\Delta_\mathrm{r} H^\ominus}{RT^2}$$

Integration gives

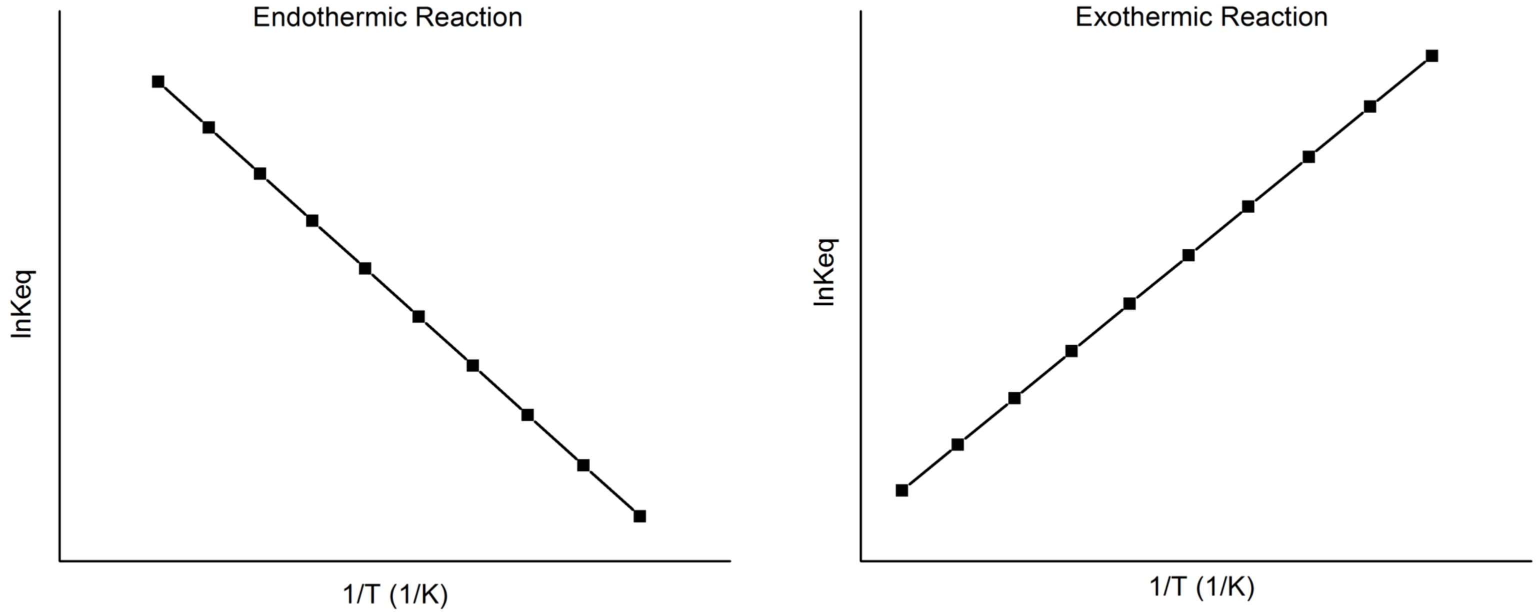
$$\int_{\ln K_1}^{\ln K_2} \mathrm{d}\ln K = \int_{T_1}^{T_2} \frac{\Delta_\mathrm{r} H^\ominus}{RT^2}\,\mathrm{d}T$$

Assuming that $\Delta_\mathrm{r} H^\ominus$ is constant,

$$\ln K_2 - \ln K_1 = \frac{\Delta_\mathrm{r} H^\ominus}{R}\int_{T_1}^{T_2} \frac{1}{T^2}\,\mathrm{d}T = -\frac{\Delta_\mathrm{r} H^\ominus}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

T-Dependence of K

$$\ln K_2 - \ln K_1 = -\frac{\Delta_r H^{\ominus}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$



(Image: https://en.wikipedia.org/wiki/Van_%27t_Hoff_equation)

Summary

Multi-component Gibbs energy

$\mathrm{d}G = \sum_i \mu_i \mathrm{d}n_i$

Chemical reaction

- Extent of reaction

  $\mathrm{d}n_i = \nu_i \mathrm{d}\xi$

Gaseous reactions

- Equilibrium constants

  K_p and K_c

- Theoretical calculation

- Reaction quotient
- Equilibrium constant

Le Chatelier's principle

- Amount change
- Volume change
- Temperature change
  - van't Hoff equation

Chemical potential

$\mu = \mu^\ominus + RT \ln a$

Activity approximations